

Physiological Indices to Predict Driver Situation Awareness in VR

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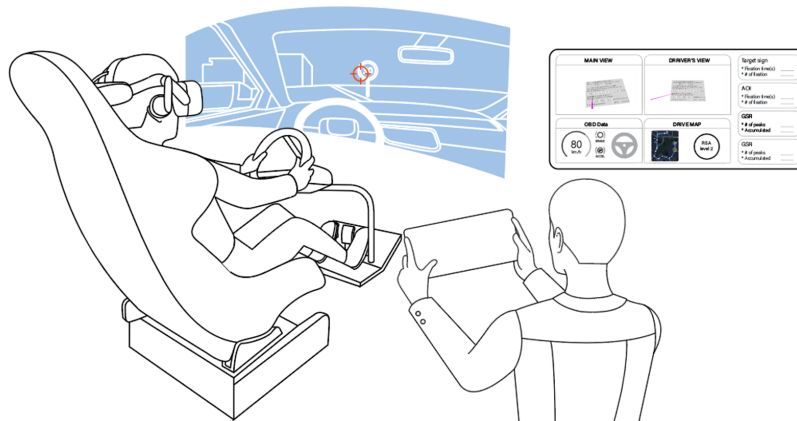


Figure 1: A schematic diagram for the sensor-based classification of driver situation awareness levels

ABSTRACT

Understanding drivers' states is essential for providing personalized interventions and adaptive feedback in vehicles, thereby ensuring safer driving and a more comfortable driver experience. As driving tasks necessitate understanding and reaction to the rapidly changing road environment and successful management of working memory to prevent dual-task interference, driver situation awareness (SA) should be primarily considered for such interventions and feedback. However, due to its complex nature, most of the current methods for measuring SA rely on questionnaires. In this work, we aim to develop sensor-based assessments of driver SA, especially towards road signs, which play an important role in immediate decision-making during driving. We collected eye-tracking data and physiological responses from 32 participants during simulated driving and annotated this data according to the levels of SA

that drivers achieved. Our ensemble-based machine learning model that uses physiological measures and gaze-related features demonstrated an accuracy of 78.02% in a three-class driver SA prediction. Since SA is a key component in evaluating vehicular interfaces, our VR-based approach holds the potential for the iterative design of in-car infotainment applications and road environments. The method also lays a foundation for SA-adaptive cooperation between human drivers and AI in the context of advanced driver assistance.

CCS CONCEPTS

• Human-centered computing → Ubiquitous computing.

KEYWORDS

eye-tracking, physiological response, driver situation awareness

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1 INTRODUCTION

Situation Awareness (SA) is the process of understanding ‘what is going on’ in the surrounding environment [8]. For safe driving, drivers must achieve SA to respond appropriately to rapidly changing environments. However, ubiquitous computers permeating vehicles can either support or hinder driver SA. While navigational aids facilitate SA attainment [27], mobile phones and other in-vehicle services cause dual-task interference [26]. As these scenarios make the interactions between the driver and the road environment increasingly complex, it has become important to understand driver SA levels for managing critical situations.

Various environmental factors, such as road signs, traffic, roadway complexity, and pedestrians, contribute to a driver’s acquisition of SA [34]. Among these, signs play an essential role in supporting decision-making at key wayfinding points. Therefore, drivers must continually interpret road sign information for immediate decision-making. However, because drivers are engaged with driving tasks, merely fixating on signs does not necessarily guarantee their awareness [6]. As a result, a driver’s awareness of road signs should be evaluated within the context of their overall situational awareness, including noticing and understanding the information. Evaluation of driver’s awareness within the SA framework will enable a more precise measurement that can distinguish between failures in understanding due to cognitive overload and missing critical information despite having glanced at the sign.

2 RELATED WORK

2.1 Situation Awareness Measurement

Endsley’s model conceptualized SA as a three-stage process [8]: Level 1 is the perception of elements in the environment. Level 2 involves the interpretation of their meanings, and Level 3 is the projection of their future state. We adapt this model in our study to encompass Level 0 (not perceived), Level 1 (perception), and Level 2/3 (comprehension and projection), following with Liang et al. [25]. We emphasize levels 1 and 2, considering that they are more significant indicators of critical situations [15, 39].

Current methods for SA measurement primarily rely on questionnaires. Endsley’s Situation Awareness Global Assessment Technique (SAGAT) [7] offers an objective approach. It decides SA based on the accuracy of responses to key information, as opposed to the subjective Likert-scale reporting of the Situation Awareness Rating Technique (SART) [38]. Given that it temporarily suspends the ongoing task to query users, thereby avoiding the acquisition of information during the survey, SAGAT is particularly suitable for simulations. For driver SA, the Daze method [28, 36] probes event perception and comprehension during driving. Our study’s ground-truth SA annotation blends SAGAT’s task suspension with Daze’s questionnaire items.

2.2 Physiological Measures of Driver States

Driver state recognition has been a topic of long interest within the UbiComp community, as it is the initial step for adaptive and personalized interventions in vehicles [11, 19, 22, 35]. While immediate measures such as the experience sampling method provide direct data [4, 35], physiological responses have also been favored due to

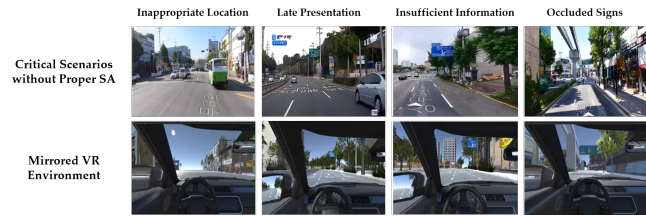


Figure 2: Sample scenarios that mirrored the real-world driving environments

the versatility across various facets of affection and cognition [12], and their potential for uninterrupted measurement [44]. Indeed, even within driving-focused studies, physiological responses have been used to predict cognitive load [29], interruptibility [19, 40], emotions [31], attention [10], stress [30], and drowsiness [33].

However, despite its importance in driving, physiological measures of driver SA levels remain a challenge to date [1]. Some studies attempted to predict binary driver SA (low, high) using EEG [23], estimate miners’ SA utilizing eye-trackers and ECG [24], and measure air traffic controller’s SA with eye-tracking [42]. However, only a few studies have conducted classifications based on the SAGAT information within the framework of Endsley’s model (e.g., aircraft operators’ SA using eye-tracking, EMG, and heart rate [5]), and none have specifically focused on predicting driver SA levels. In this study, we classified sign-related driver SA levels based on Endsley’s model, utilizing physiological and gaze-related measures indicative of cognitive states and attention, recognizing that SA is primarily a visual and cognitive construct [32].

3 MATERIALS AND METHODS

3.1 Implementation

3.1.1 Driving Simulator and Physiological Measures. We developed a driving simulation with motion feedback to gather driver SA data from human subjects during VR driving (Figure 1). The system incorporates an HTC VIVE Pro Eye Head-Mounted Display (HMD) (2880×1600px, 110 degrees Field of View, 90Hz) and an A3 motion simulator. We chose an HMD-based environment to ensure reliable measurement of gaze-related measures without the impact of external light exposure or tracking failure due to the driver’s head movement. Eye-tracking data were collected at 120Hz with 0.5°–1.1° accuracy and a 110° field-of-view. Galvanic skin response (GSR) was measured on a participant’s shoulder—which has the least motion artifact during driving—at a sampling rate of 500Hz, using the Cognionics AIM Physiological Monitor.

3.1.2 Driving Scenarios. Given that most standard driving scenarios may not include failure in SA, we concentrated on incorporating critical scenarios where drivers struggle to achieve appropriate SA due to inadequate signage. We used online search engines and video platforms like YouTube to understand the precise environments and sign designs of scenarios where complaints were made or accidents occurred. We categorized the sign-related critical scenarios into four groups: inappropriate location, late presentation, insufficient information, and occluded signs. After identifying these scenarios,

we replicated the environments within a VR setting and created 15 driving scenarios (Figure 2).

3.1.3 Situation Awareness Probes. We included a VR survey for each session to assess driver SA, serving as a ground truth for the sensor data. The survey appears when the participant stops in front of a red light after each session, pausing the driving simulation. Based upon the Daze method [28, 36], the survey asks whether they noticed the sign (perception, dividing levels 0-1) and about the direction of their destination (comprehension, dividing levels 1-2). To prevent participants from overly focusing on signs to answer the questions accurately, we included dummy items for both level 1 and level 2 information (e.g., the number of people crossing roadways).

3.2 Experiment

3.2.1 Procedure. In the beginning, participants were explained about the experiment, wore physiological sensors, and completed demographic questions and our modified visual activity questionnaire [37]; measured Q3, 6, 16, 20, 24 for visual search problems and Q8, 13, 18, 22, 30, 33 for visual processing problems. They then underwent a training session to get familiar with the driving environment and learned how to respond to the survey in VR. The actual experiment began once they felt sufficiently comfortable with the system. Before each driving session, participants were informed of their destination. During the experiment, participants were instructed to drive the vehicle to the destination using information from signs on roads and navigational aids while maintaining their lane, adhering to traffic signals, and under a speed limit of 60km/h. Participants drove through 15 different scenarios, and the driving scenarios' order was counterbalanced to mitigate learning effects.

3.2.2 Participant. We recruited 32 participants (female: 10, age: $M = 24.78$, $SD = 4.97$) with a minimum of less than one year and a maximum of more than five years of driving experience. (less than 1 year: 11, 1-3 years: 11, 3-5 years: 5, 5+ years: 5). Participants showed no issues with visual search ($M = 2.35$, $SD = 0.17$) and visual processing speed ($M = 2.11$, $SD = 0.15$), as assessed by our modified VAQ in a 5-point Likert scale.

4 RESULTS

4.1 User Evaluation on Driving Scenarios

To assess our platform's feasibility as a tool for sign-related driver SA measurement, we examined the difficulty of the scenario and simulator sickness using a 5-point Likert scale. The scenario was perceived to have moderate driving difficulty ($M = 2.19$, $SD = 1.23$), suggesting it wouldn't overly burden the sign recognition task with the excessive driving workload. Also, the VR driving did not induce significant simulator sickness that could potentially impact physiological responses ($M = 1.85$; $SD = 0.88$).

4.2 Dataset Composition

We obtained a dataset comprising 480 complete sessions (32 participants $\times$ 15 scenarios) with eye-tracking data, physiological responses, and driving performance measures, which were annotated with the sign-related driver SA using the SAGAT-based method. 30.2% of total data were labeled as level 0, 25.1% as level 1, and 44.7% of total data were annotated as level 2.

4.3 Situation Awareness Measurement

4.3.1 Feature Extraction. To exclude the effects of cognitive load the survey induces, we included only data from the 1-minute intervals before and after passing each target sign for analysis. Each session's data was preprocessed into 34 physiological and gaze-related features. These include six physiological measures: average pupil size (left and right), eye blinks (left and right), average GSR, and the number of GSR peaks. 27 Gaze-related features comprising gaze visits, total fixation time, and first glance time on target wayfinding sign, target information on the wayfinding sign, roads, pedestrians, traffic signs, informational signs, mirrors, navigational aids, and other elements were also considered. The ratio of target visits divided by total visits was included as another gaze feature. We split the dataset into an 8:2 ratio for training and testing.

4.3.2 Model Description. We evaluated various classification models, including k-nearest neighbors (K-NN), multi-layer perceptron (MLP), support vector machine (SVM), logistic regression classifier (LRC), gradient boosting regression trees (GBR Trees), extra trees, and random forest, along with their ensembles. For ensemble models, we tested both soft voting, which sums up the probability from each model, and stacking models that employ a meta-classifier. We implemented a grid search to find the optimal hyperparameters with the highest accuracy for each individual model and for all possible ensemble combinations of individual models. The model training process used 10-fold cross-validation for model validation and hyperparameter tuning. For each ensemble model's final classifier, we used random forest, extra tree, and logistic regression for the voting classifier, while random forest, extra tree, and MLP models were employed for stacking classifiers. Notably, the extra tree classifier as the final estimator showed the highest accuracy for both ensemble techniques.

4.3.3 Model Performance. The stacking model achieved the highest accuracy of 78.02% when trained and tested with all the features, followed by the soft voting model, which recorded 75.83% accuracy (Table 1). We also calculated precision, recall, area under the receiver operating characteristic curve (AUC), and F1-score for the models using all features (Table 2). We present the macro-average results for each measure to emphasize the models' performance for each class rather than each individual data point. The stacking model utilizing all features achieved the highest F1-score of 72.49, which, being lower than its accuracy, suggests an imbalance in performance across the classes.

5 DISCUSSION

5.1 Model Performance Enhancement

While our experiment concentrates solely on the sign-related driver SA, our stacking model achieved an accuracy of 78.02%, which is comparable to the EEG-based method [23] (74.32% accuracy) and the gaze-based study [45] (72.4% accuracy), though the goals and dataset are different. However, the f1-score is lower than the accuracy, indicating an imbalance among classes (Table 2).

Our next steps will involve increasing the dataset size, especially for SA Level 1, applying techniques to address class imbalance, and utilizing additional sensors to gather more comprehensive information about drivers' cognitive states. The ROC curve for the stacking

Table 1: Top-1 accuracy result of situation awareness level classification with physiological responses and gaze-related features

Model	Physiological responses	Gaze features	All features
K-NN	42.39	57.61	59.78
MLP	41.76	50.55	67.03
SVM	45.05	57.14	67.03
LRC	48.35	58.14	68.13
GBR Tree	41.76	62.64	69.23
Extra Tree	45.05	61.54	73.63
Random Forest	44.35	62.64	74.63
Soft Voting	45.05	60.64	75.82
Stacking	48.03	63.06	78.02

Table 2: Precision, recall, AUC, and f-1 score of all-features models

Model	Precision	Recall	AUC	F1-score
K-NN	56.40	56.96	73.94	53.60
MLP	64.1	65.02	76.76	64.30
SVM	61.78	62.06	76.49	61.66
LRC	57.94	59.37	76.94	58.13
GBR Tree	63.69	64.10	77.75	62.17
Extra Tree	74.12	67.34	81.10	62.60
Random Forest	71.70	72.64	79.80	69.01
Soft Voting	73.07	73.13	80.49	67.85
Stacking	78.82	78.41	81.14	72.49

model indicates a relative weakness in accurately classifying SA level 1, while demonstrating better results for levels 0 and 2 (Figure 3). We attribute this primarily to the imbalanced proportion of classes in our dataset (level 0: 30.2%, level 1: 25.1%, level 2: 44.7%). Collecting more data for SA level 1 may enhance the model’s accuracy for these cases, leading to an overall improvement in the model’s performance.

Another contributing factor could be the subtle differences within level 1. An individual’s capacity to comprehend a sign can vary based on their working memory, driving experience, sign familiarity, and in-situ factors such as cognitive load and attention. Consequently, eye-tracking or physiological responses may differ for class 1 (“seeing but not comprehending”), making the distinction between this class and the other two more subtle. To improve the accuracy in classifying driver SA, the use of additional sensors such as EEGs could be considered; it can provide a more direct measure of cognitive processes. However, this may increase the complexity of both the experimental setup and data analysis to the current real-time measurement system. The decision thus should be tailored to individual purposes and application requirements, balancing the potential trade-offs between accuracy and speed.

5.2 Potential Applications of Our Method

Our system can be adapted for the objective assessment of signs, road infrastructures, and other interfaces. In addition to the gaze measures for driver attention [14, 43], our approach provides SA information concerning driver perception and comprehension. By

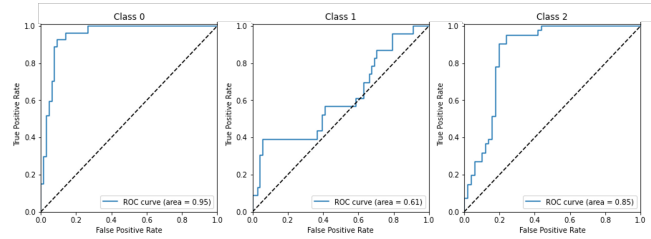


Figure 3: ROC curve for each class of the stacking model

altering different designs of road elements they can be iteratively designed from the driver SA viewpoint for safer road conditions. Similarly, the impact of dashboard [21] and windshield designs [13, 20], or other interfaces [3] on driver SA can be evaluated. Given VR’s effectiveness in automotive UI/UX test [16] and the validity of driving simulators [41], this can be a scalable and reliable method.

Our study serves as a basis for AI systems to understand a driver SA, facilitating effective human-AI interaction [17] and collaboration [9]. As SA is an important aspect of information delivery in automated vehicles, our method can be utilized for in-vehicle explanation services [18], to foster trust with SA-adaptive feedback. It can also be used in AI systems to determine suitable moments for in-vehicle intervention [4, 40] to enhance the overall driving experience. Moreover, it opens possibilities for human-AI cooperation for safer and more efficient driving, where control can be shifted based on the driver’s SA [2]. Our research thus contributes to the evolving field of human-AI interaction in vehicular environments.

6 CONCLUSION

In this study, we constructed simulated driving environments, replicating sign-related scenarios known to challenge drivers in achieving SA. We conducted experiments involving 32 participants, gathering physiological responses during driving, annotated with SAGAT-based ground truth data. Our machine learning approach predicted driver SA levels pertaining to road signs from these physiological responses, achieving 78.02% accuracy in three-level classification. While our method leverages both arithmetic and contextual features specific to driving scenarios, it can be scaled to on-road environments with the appropriate algorithms. Given that the classification levels are based on Endsley’s model, our method enables AI systems to predict driver SA, regarding their perception and comprehension with the environment. This expands our study’s potential applications to the drivers’ interaction with the environments, interfaces, and AI-infused systems.

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